Holes: Layout Algorithms

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1 Introduction

This document describes the layout algorithms implemented in the Holes project, which was used to run simulations reported in the articles "Size Matters: Optimal Test Pit Size for Dispersed Archaeological Test Excavations" and "Test Pit Layout and Site Geometry: Optimising Dispersed Archaeological Test Excavation Programs" - referred to here as "Layout" - by Oakes and McLaren. See those articles for background. We simulate archaeological digs to explore success of different test pit sizes, numbers of test pits, and layout strategies, with different sizes, shapes, and types of archaeological sites. Our simulations take place on a square field of dimension $n \times n$ (typically $100m \times 100m$). The test pit sizes for simulations used in the "Size Matters" article are $0.5m \times 0.5m$ (small pits) and $1m \times 1m$ (large pits). Only small $0.5m \times 0.5m$ pits are used in "Layout". An archaeological site is placed randomly on the field, and pits dug according to a layout algorithm to see if they find the site (or find individual artefacts in some experiments conducted with real world data in "Size Matters").

2 Hexagonal-Like Layout Algorithm

This is the only algorithm used for the simulations reported in "Size Matters". In "Layout" we explore other layout algorithms (described below) as well as this one. We define a staggered grid as one consisting of several straight and parallel rows of test pits, with the distance betwen adjacent pits in a row being equal, the distance between rows being equal, and with each second row indented (staggered) by half the distance between adjacent pits in a row. In addition we enforce exact borders around the edges of the field where no pits are dug. The reason for fixing borders is to avoid anomalous results when a pit is close to the edge of the field (noting that the archaeological site is placed so that at least two thirds of at least two sides of its bounding box is within the field) or the distance from the field edge to the nearest pit is too great. The left and right borders are a proportion of the distance between pits in a row, and the top and bottom borders are a proportion of the distance between rows (a proportion of 1/4 was used in the experiments). Note that the centre of a pit may fall on a

border. See figure 1. For a given total number of pits h we need to decide how many pits will be dug in our field horizontally (h_x), and how many vertically (h_y).

First some definitions:

- h is the total number of pits,
- h_x and h_y are the number of pits along the x axis and the y axis of the field respectively,
- n_x and n_y are the width and height of the field respectively (we only consider square fields in the experiments),
- a, b, and c are distances between pits as shown in figure 1,
- $1/p$ is the proportion of a and c used for the borders around the edges of the field (horizontally and vertically respectively),
- $pitSize$ is the length (and width) of the square test pits.

A staggered layout is hexagonal when a and b shown in figure 1 are equal. A hexagonal layout is considered optimal in certain circumstances where the archaelogical sites are circular. In the hexagonal-like algorithm we choose values for h_x and h_y that make the layout almost hexagonal (while keeping the borders around the edges of the field as described above). In section 3 we vary the borders to create a truly hexagonal layout. In experiments the hexagonal-like and hexagonal algorithms had similar success rates except for some site orientations and elongations where the hexagonal-like algorithm performed slightly better.

In this section, except where otherwise apparent, we treat pits as points, ignoring $pitSize$. So the centre of a pit may fall on the border. We assume $pitSize$ is much less than the size of the field.

We now derive some equations for h_x and h_y, starting with some equations evident from figure 1:

$$h = h_x \cdot h_y \tag{1}$$

$$n_x = \frac{2a}{p} + \left(h_x - \frac{1}{2}\right) a \tag{2}$$

$$= a\left(\frac{2}{p} + h_x - \frac{1}{2}\right) \tag{3}$$

$$n_y = \frac{2c}{p} + (h_y - 1)\, c \tag{4}$$

$$= c\left(\frac{2}{p} + h_y - 1\right) \tag{5}$$

$$a^2 = \left(\frac{a}{2}\right)^2 + c^2 \qquad \text{Pythagorean Theorem} \tag{6}$$

We write a formula for a in terms of n_x, h_x, and p:

$$a = \frac{n_x}{\frac{2}{p} + h_x - \frac{1}{2}} \qquad \text{from 3} \tag{7}$$

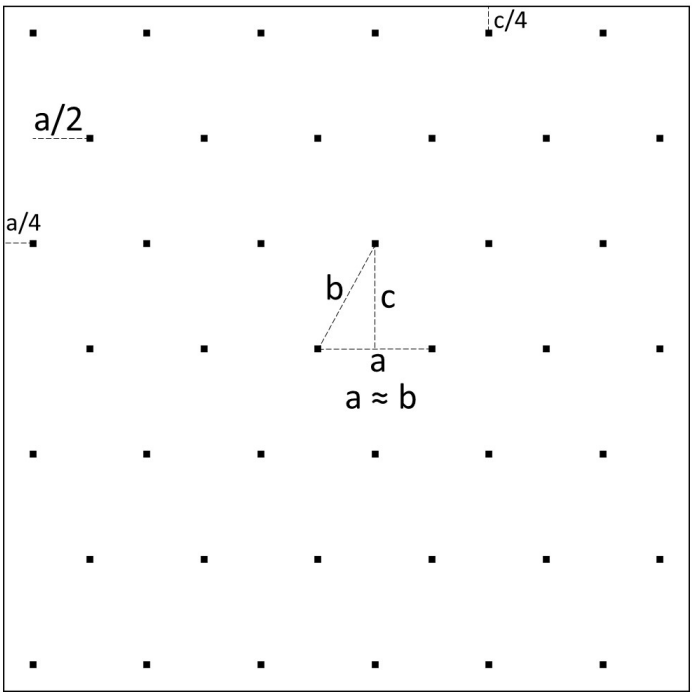


Figure 1: Hexagonal-like grid for $h = 42$ test pits and $p = 4$. A staggered layout is hexagonal when $a = b$.

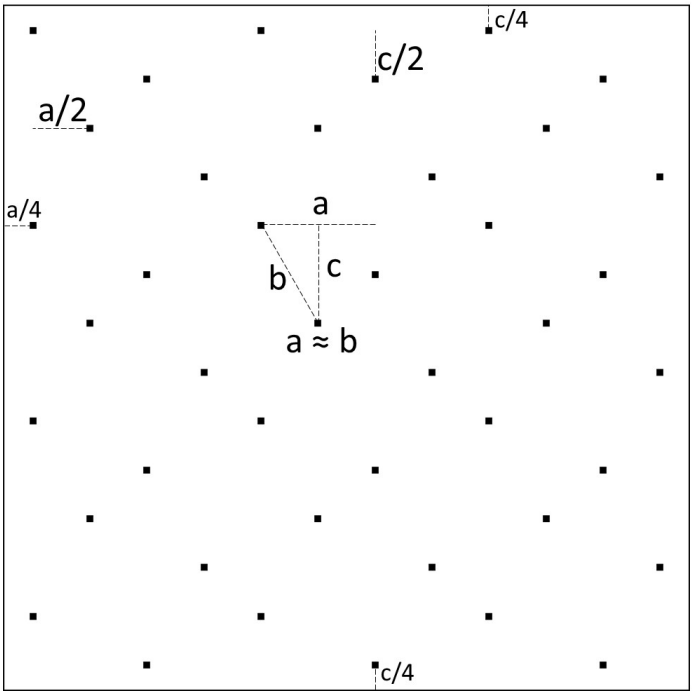


Figure 2: StaggerXY grid for $h = 42$ test pits and $p = 4$.

Now we write a formula for a in terms of n_y and h_y and p:

$$\begin{aligned} a^2 &= \frac{a^2}{4} + c^2 \qquad \text{from 6} && (8)\\ \frac{3}{4}a^2 &= c^2 && (9)\\ a &= \frac{2c}{\sqrt{3}} && (10)\\ &= \frac{2n_y}{\sqrt{3}\left(\frac{2}{p} + h_y - 1\right)} \qquad \text{from 5} && (11) \end{aligned}$$

We equate the two formulae for a and find a quadratic equation for h_x:

$$\begin{aligned} \frac{n_x}{\frac{2}{p} + h_x - \frac{1}{2}} &= \frac{2n_y}{\sqrt{3}\left(\frac{2}{p} + h_y - 1\right)} && (12)\\ \sqrt{3}n_x\left(\frac{2}{p} + h_y - 1\right) &= 2n_y\left(\frac{2}{p} + h_x - \frac{1}{2}\right) && (13)\\ & && (14) \end{aligned}$$

Multiplying by h_x gives us:

$$\sqrt{3}n_x\left(\frac{2h_x}{p} + h - h_x\right) = 2n_y\left(\frac{2h_x}{p} + h_x^2 - \frac{h_x}{2}\right) \tag{15}$$

$$2n_y h_x^2 + \left(\frac{4n_y}{p} - n_y - \frac{2\sqrt{3}n_x}{p} + \sqrt{3}n_x\right)h_x - \sqrt{3}n_x h = 0 \tag{16}$$

$$\tag{17}$$

which, for $p = 4$ simplifies to:

$$2n_y h_x^2 + \frac{\sqrt{3}n_x}{2}h_x - \sqrt{3}n_x h = 0 \tag{18}$$

We then solve for h_x using the quadratic formula and calculate:

$$h_y = \frac{h}{h_x} \tag{19}$$

We round the values to get integer values for h_x and h_y (this, while necessary, is where the algorithm diverges from the truly hexagonal layout captured in the equations) and recalculate a and c to reflect the rounded values:

$$\begin{aligned} c' &= \frac{n_y}{\frac{2}{p} + round(h_y) - 1} && (20)\\ a' &= \frac{n_x}{\frac{2}{p} + round(h_x) - \frac{1}{2}} && (21)\\ border_x &= \frac{a'}{p} && (22)\\ border_y &= \frac{c'}{p} && (23) \end{aligned}$$

Then we layout the pits with the given borders, so that $round(h_x)$ pits in a row are a' apart, $round(h_y)$ rows are c' apart, and every second row is indented (staggered) by $a'/2$. Note again that p was set to 4 for the experiments.

If $border_x$ or $border_y$ is less than $pitSize/2$ the layout algorithm fails (note a pit location is specified by the coordinates of its centre). However if desired it is possible to set the borders to $pitSize/2$ and recalculate h_x and h_y using similar maths to the above. This is not currently implemented in the code.

3 Hexagonal Layout Algorithm

This section describes one way to implement a hexagonal layout algorithm. We do not claim that it is the best way. We start with the equations for a hexagonal-like layout with a horizontal border of a/p and vertical border of c/p, calculating h_x and h_y as in equations 16 and 19. Then we vary the borders so that it is exactly hexagonal. First fix $border_y$ and c to their original values and change $border_x$ and a:

$$c' = \frac{n_y}{\frac{2}{p} + round(h_y) - 1} \tag{24}$$

$$a' = \frac{2c'}{\sqrt{3}} \qquad \text{from 10} \tag{25}$$

$$border_x = \frac{n_x - a'(round(h_x) - \frac{1}{2})}{2} \tag{26}$$

$$border_y = \frac{c'}{2} \tag{27}$$

If $border_x < pitSize/2$, the pits won't fit (pits are specified by the coordinates of their centre). In that case we set $border_x$ it to its minimum value of $pitSize/2$ and vary the other values as follows:

$$a'' = \frac{n_x - pitSize}{round(h_x) - \frac{1}{2}} \tag{28}$$

$$c'' = \frac{\sqrt{3}a''}{2} \qquad \text{from 10} \tag{29}$$

$$border_x = \frac{pitSize}{2} \tag{30}$$

$$border_y = \frac{n_y - c''(round(h_y) - 1)}{2} \tag{31}$$

Our value for a may now be a' or a'', so we just refer to it as a. Likewise for c. If $a < pitSize$, $border_x < pitSize/2$, $c < pitSize$, or $border_y < pitSize/2$, we are unable to create a hexagonal layout using this method. Otherwise we layout the pits with the given borders, so that $round(h_x)$ pits in a row are a apart, $round(h_y)$ rows are c apart, and every second row is indented (staggered) by $a/2$. This algorithm was not used in the articles except to test that the hexagonal-like and hexagonal algorithms are similar in performance.

4 StaggerXY Layout Algorithm

The hexagonal-like and hexagonal layout algorithms result in a particular type of staggered grid. This staggering is in the horizontal direction, with every second row indented. We can also stagger in the vertical direction, so that every second pit in each row is shifted down (see figure 2). This improves on the hexagonal-like layout in some cases when the archaeological site becomes elongated rather than being circular. Adapting the hexagonal-like layout algorithm to do this requires a slight difference to the maths used to calculate h_x and h_y. The difference begins with a difference to equation 5, namely

$$n_y = c\left(\frac{2}{p} + h_y - \frac{1}{2}\right) \tag{32}$$

Then similar maths to section 2 gives

$$2n_y h_x^2 + \left(\frac{4n_y}{p} - n_y + \frac{\sqrt{3}n_x}{2} - \frac{2\sqrt{3}n_x}{p}\right) h_x - \sqrt{3}n_x h = 0 \tag{33}$$

which for $p = 4$ simplifies to:

$$2n_y h_x^2 - \sqrt{3}n_x h = 0 \tag{34}$$

This can be solved using the quadratic formula for equation 33 or simply for equation 34. Then we calculate:

$$h_y = \frac{h}{h_x} \tag{35}$$

$$c' = \frac{n_y}{\frac{2}{p} + round(h_y) - \frac{1}{2}} \tag{36}$$

$$a' = \frac{n_x}{\frac{2}{p} + round(h_x) - \frac{1}{2}} \tag{37}$$

$$border_x = \frac{a'}{p} \tag{38}$$

$$border_y = \frac{c'}{p} \tag{39}$$

Then we layout the pits with the given borders, so that $round(h_x)$ pits in a row are a' apart, $round(h_y)$ rows are c' apart, every second row is indented (staggered) by $a'/2$, and every second pit in each row is shifted down by $c'/2$. Note that due to the vertical staggering the layout is no longer hexagonal (or hexagonal-like).

5 Random Layout Algorithm

The performance of the systematic algorithms above declines rapidly as site elongation ratio increases, as the elongated site can sit between the regularly

spaced test pits. To overcome this, we consider the random and halton layout algorithms which perform comparatively well at larger site elongation ratios. The random layout algorithm chooses pit locations according to a pseudo-random number generator (of course conventional computers cannot produce truly random results!). The pits do not overlap. Margins are as for a square grid with the same number of pits with margins of $1/p$ the distance between pits (and rows), i.e. $\frac{n_x}{p\left(\sqrt{h}-1+\frac{2}{p}\right)}$, noting that the field is square and ignoring the fact that h may not be a square number.

6 Halton Layout Algorithm

The Halton layout algorithm lays out the pits according to a 2-dimensional scrambled Halton sequence, which appears random but does not cluster pits close to each other (which may happen in the random layout). See [Owen, A.B. (2017). A randomized Halton algorithm in R. arXiv:1706.02808] for the details of a scrambled halton sequence, which algorithm is implemented in the SciPy python library [Virtanen P. et al. (2020). SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. Nature Methods 17(3), 261-272. DOI: 10.1038/s41592-019-0686-2]. Margins are as for the random layout (above).

7 RandomisedStaggerXY Layout Algorithm

While the random and halton layout algorithms perform better than the hexagonal-like and staggerXY algorithms for larger site elongation ratios, they do not perform as well as those algorithms for smaller site elongation ratios. So we consider a hybrid algorithm. We start with the staggerXY algorithm and then randomise the positions of the pits in both x and y directions. The position of each pit (first the x position and then the y position) is randomised according to a normal distribution with the original position as mean, so it is more likely to land close to its original position, with the probability declining as distance from the original position increases. The standard deviation in the x direction is one eighth the horizontal distance between pits, and in the y direction is one eighth the vertical distance between rows. The amount by which it is randomised is capped at half the distance between pits in a row (for adjustment in the x direction) and half the distance between rows (for adjustment in the y direction). In addition the borders of the staggerXY algorithm are maintained for consistency, so pits on the left and right borders are not randomised in the x direction and pits on the top and bottom borders are not randomised in the y direction. Other standard deviations are possible, but this one worked well in experiments, providing a balance between performance at small and larger site elongation ratios.